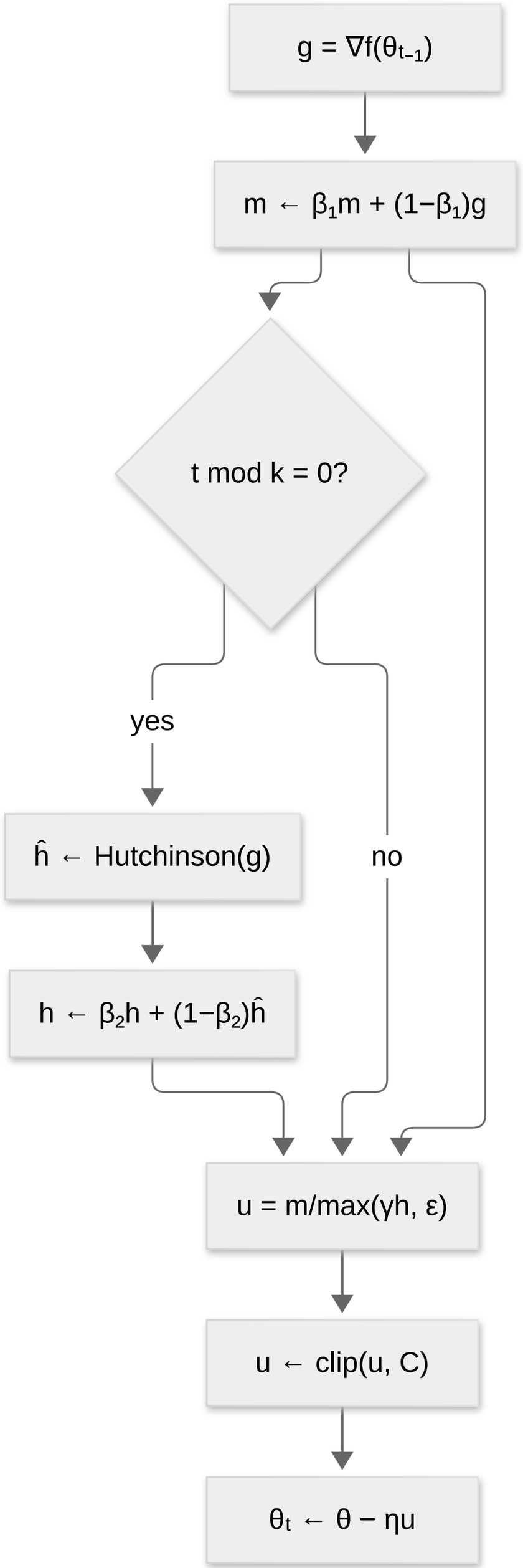

$g = \nabla f(\theta_{t-1})$
$m \leftarrow \beta_1 m + (1-\beta_1) g$
t mod k = 0?
yes
no
$\hat{h} \leftarrow$ Hutchinson(g)
$h \leftarrow \beta_2 h + (1-\beta_2)\hat{h}$
$u = m/\max(\gamma h, \varepsilon)$
$u \leftarrow \text{clip}(u, C)$
$\theta_t \leftarrow \theta - \eta u$